



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

AI Video Intelligence Platform – An AI-Powered System

The domain of the Project: Artificial Intelligence — Information Retrieval & Natural Language Processing, applied to Indian e-Governance (Government Scheme Discovery)

COURSE NAME: AI&ML

Team Mentors (and their designation):

Mrs. Rajalakshmi C

Team Members:

Mr. Kethan pawar

Period of the project

March 2026 to September 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “AI Video Intelligence Platform – An AI-Powered System for Understanding, Searching and Extracting Knowledge from Video Content” has been mentored by Rajalakshmi C, organised by SURE Trust, from April 2026 to August 2026, for the benefit of educated unemployed rural youth, providing them with hands-on experience in working on industry-relevant projects that would take them closer to prospective employers. I declare that, to the best of my knowledge, the members of the team mentioned below have worked on the project successfully and enhanced their practical knowledge in the domain of Artificial Intelligence and Machine Learning.

Team Members:

Mr. Kethan pawar

Mentor's Name
Bhagya Bedke

Mentor's Name
Rajalakshmi C

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Table of contents

1. Executive Summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. System Architecture
6. AI/ML Technologies Used
7. Project Modules and Services
8. Social / Industry Relevance
9. Learning & Reflection
10. Limitations
11. Future Scope
12. Conclusion



Executive Summary

Project Overview

1.1 Project Overview

The **AI Video Intelligence Platform** is an AI-powered application designed to automatically understand and extract meaningful information from video content.

Modern videos contain multiple types of information simultaneously. A video may contain spoken conversations, multiple speakers, text displayed on screens, people, objects, activities and other visual information. Traditionally, users need to watch the complete video to find a particular piece of information.

The proposed system reduces this effort by processing the different information channels of a video using specialized Artificial Intelligence and Machine Learning technologies.

The system processes:

- Video
- Audio
- Speech
- Speakers
- On-screen text
- Visual information
- Timestamps

These different outputs are then combined to provide useful video intelligence such as:

- Timestamped transcripts
- Speaker-aware transcripts
- OCR results
- Video summaries
- Key points
- Topics
- Important moments
- Searchable information
- Video-based question answering
- Educational content
- Analytics
- Exportable reports

The application is implemented using **Python and Streamlit** with specialized AI/ML components including **Faster-Whisper, Pyannote Audio, PaddleOCR and computer vision techniques**.



1.2 Problem Statement

Videos are becoming one of the most common sources of information in education, business, meetings, training, surveillance, media and online learning.

However, finding specific information inside long videos is difficult.

For example:

- A student may want to find a particular topic from a 2-hour lecture.
- A company may want to find an important statement from a meeting.
- A user may want to know what text appeared on a presentation slide.
- A researcher may need to identify when a particular speaker discussed a topic.
- A teacher may want to convert a lecture into notes and questions.

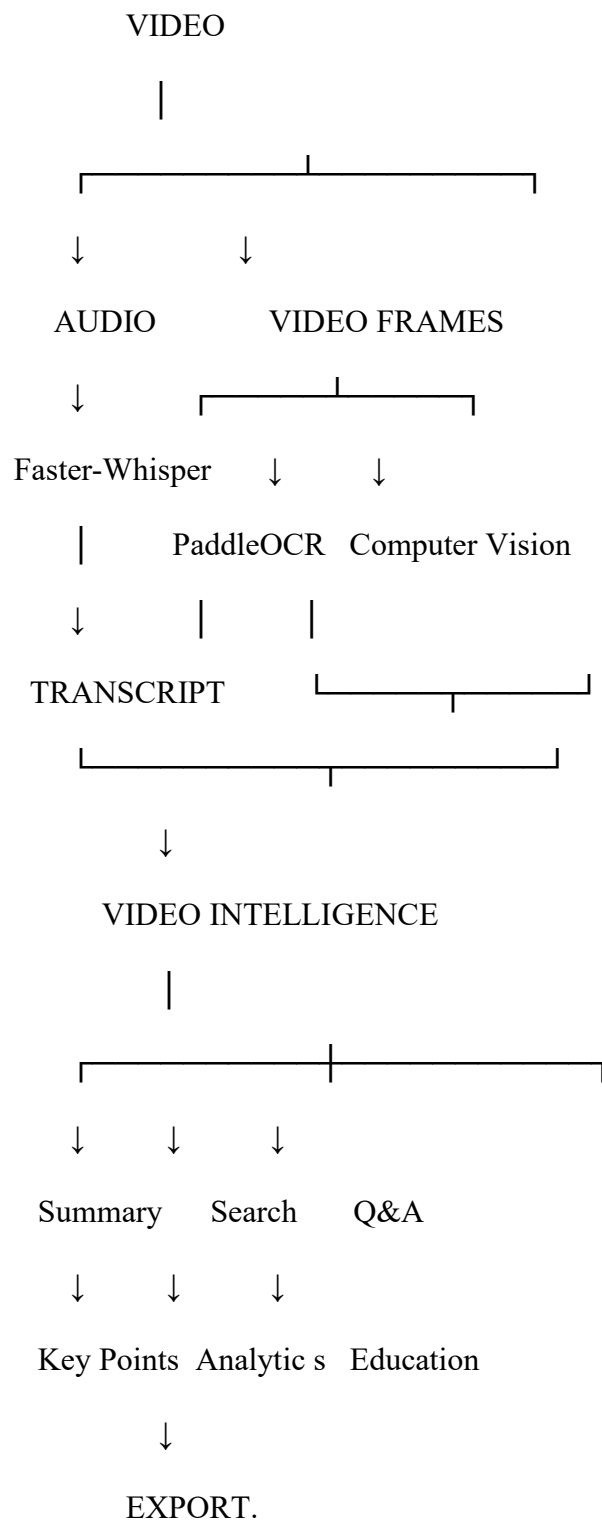
Normally, users have to watch the complete video or manually search through it.

Therefore, there is a need for a system that can **automatically process video and convert unstructured video content into searchable and useful information.**



1.3 Proposed Solution

The AI Video Intelligence Platform solves this problem by creating a multimodal video-processing pipeline





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

2. INTRODUCTION

2.1 Background

Video is a rich source of information because it contains multiple modalities at the same time.

A single educational video, for example, may contain:

- Teacher speech
- Student questions
- Presentation slides
- Text displayed on screen
- Diagrams
- People
- Objects
- Important events

A conventional video player mainly provides playback functionality. It does not automatically understand all these information sources.

The AI Video Intelligence Platform aims to transform video from a passive media format into an **intelligent and searchable information source**.

2.2 Need for the Project

The main motivation for this project is the difficulty of manually analyzing long videos.

Consider a 3–5 hour lecture or meeting.

A user searching for one particular statement might have to spend several hours watching the video.

The proposed system can instead:

1. Process the video.
2. Extract its audio.
3. Convert speech into text.
4. Identify different speakers.
5. Detect visible text.
6. Analyze visual information.
7. Identify important moments.
8. Generate useful insights.
9. Allow the user to ask questions about the video.

This significantly improves the accessibility and usability of video information.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

3. PROJECT OBJECTIVES

Objective 1 — Automated Video Processing

Develop a system that accepts a video and automatically extracts relevant information.

The system processes:

- Video metadata
- Audio
- Frames
- Speech
- Text
- Visual information

Objective 2 — Speech Recognition

Use **Faster-Whisper** to convert spoken audio into text.

The system provides:

- Transcription
- Language detection
- Timestamped speech segments
- Searchable transcript

Objective 3— Text Detection from Video

Use Paddle-OCR to extract text visible in video frames.

Examples include:

- Presentation slides
- Titles
- Labels
- Screen text
- Captions
- Visible written information

The system stores the detected text together with confidence information and time stamps.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Objective 5 — Computer Vision

Analyze visual information present in the video.

The computer vision component can be used to understand:

- People
- Objects
- Scenes
- Visual activities

This provides information that cannot be obtained from audio alone.

Objective 6 — Multimodal Video Understanding

Combine different information sources:

Speech
+
Speaker Information
+
OCR
+
Visual Information
+
Timestamps

to create a more complete representation of the video.

Objective 7 — Video Search and Insights

Allow users to find useful information without watching the complete video.

The platform provides:

- Key points
- Topics
- Summary
- Important moments
- Searchable information
- Timestamp references

Objective 8 — Ask Your Video

Provide a natural-language question-answering interface where users can ask questions about the analyzed video.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Example:

“What is the main topic discussed in the video?”

or

“What did the speaker explain about machine learning?”

Objective 9 — Education Intelligence

Convert video content into educational material such as:

- Lecture notes
- Summaries
- MCQs
- Quizzes
- Flashcards
- Study material

Objective 10 — Reporting and Export

Allow users to export useful information in structured formats such as:

- PDF
- JSON
- CSV

4. METHODOLOGY & RESULTS

4.1 Video Upload and Validation

The user first uploads a video through the Stream-lit interface.

The system validates the uploaded file and extracts meta-data such as:

- Duration
- Resolution
- FPS
- Frame count
- Video format

The uploaded video is then assigned an internal identifier for processing.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

4.2 Frame Extraction

Video frames are extracted at configured intervals.

For example:

```
Video
↓
Frame 1
Frame 2
Frame 3
...
Frame N
```

These frames are used for:

- OCR
- Computer vision
- Visual analysis
- Important moment detection

This avoids the need to process every frame unnecessarily.

4.3 Audio Extraction

The audio track is extracted from the uploaded video.

The extracted audio is converted into a suitable format for speech processing.

The system uses:

- 16 kHz sample rate
- Mono audio
- WAV format

The extracted audio is then passed to the speech recognition module.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

5. SPEECH INTELLIGENCE

5.1 *Faster-Whisper*

The project uses **Faster-Whisper** for Automatic Speech Recognition.

Its primary task is:

Human Speech → Text

The system generates time-stamped segments.

Example:

00:00 - 00:06

"Today we will discuss machine learning."

00:07 - 00:15

"Machine learning is a subset of artificial intelligence."

5.2 *Advantages*

Faster-Whisper provides:

- Speech-to-text conversion
- Timestamped transcription
- Language detection
- GPU acceleration
- Efficient inference

The project was configured to use GPU processing where available.

6. OCR INTELLIGENCE

6.1 *PaddleOCR*

The project uses **PaddleOCR** for Optical Character Recognition.

OCR converts visible text inside an image or video frame into machine-readable text.

Video Frame



PaddleOCR



Detected Text



Confidence + Timestamp



7.2 OCR Results

The platform can provide:

- Detected text
- Confidence score
- Bounding box
- Frame information
- Timestamp

This makes video text searchable and useful for analysis.

8.COMPUTER VISION

Computer vision is used to analyze information that cannot be obtained from speech or OCR.

The vision pipeline processes video frames and extracts visual information.

Examples:

- People
- Objects
- Scenes
- Visual activities

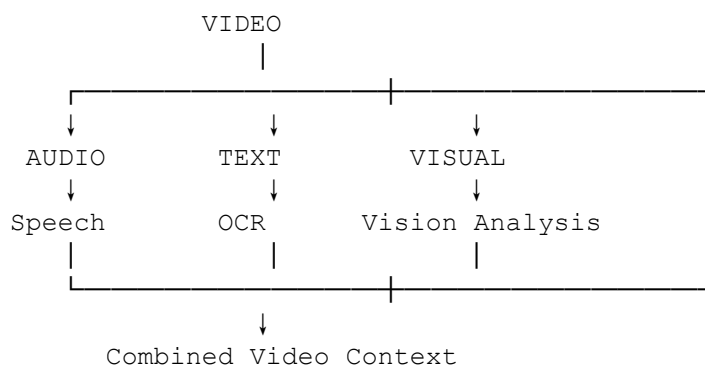
The combination of vision and OCR allows the system to understand both **what is visible** and **what is written**.

9. MULTIMODAL VIDEO UNDERSTANDING

One of the major features of the project is the combination of different information sources.

A video is not treated as only a sequence of images or only an audio recording.

The platform combines:



This provides a more complete understanding of the video.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

11. KEY POINTS AND TOPICS

The system extracts important information from the video.

It can identify:

- Main topics
- Key points
- Important concepts
- Significant discussion areas

This is particularly useful for educational videos and meetings.

12. IMPORTANT MOMENTS

The platform identifies important portions of the video and associates them with timestamps.

Example:

00:02:15 → Introduction
00:10:42 → Main concept explained
00:25:18 → Important example
00:41:05 → Conclusion

This allows users to directly navigate to relevant portions of the video.

13. ASK YOUR VIDEO

13.1 Question Answering

The platform provides an **Ask Your Video** feature.

Users can ask questions about the analyzed content.

Examples:

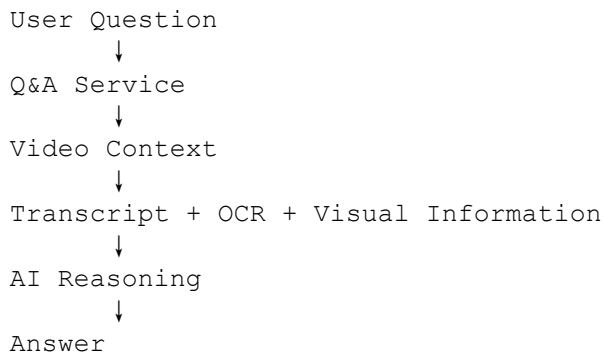
- What is this video about?
- What are the main points?
- What did the speaker explain?
- What text appeared on the screen?
- Explain the main topic.
- What happened at a particular timestamp?

The system uses the available video context to generate an answer.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

13.2 Architecture



The system can also provide source/time stamp information where available.

14. EDUCATION INTELLIGENCE

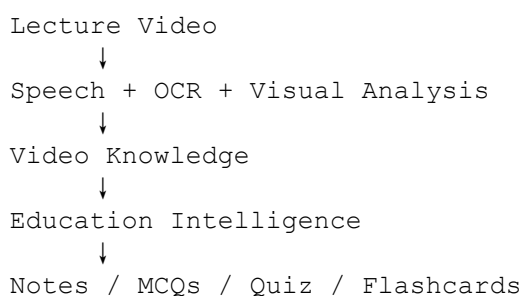
One important application of the platform is education.

The system can transform lecture videos into structured study material.

Generated resources include:

- Lecture notes
- Summary
- Key points
- MCQs
- Quiz questions
- Flashcards
- Study material

Workflow



This can reduce the effort required for students to manually create study material from long lectures.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

15. SEARCH AND ANALYTICS

The platform provides search and analytical capabilities over the processed video information.

Users can search extracted information instead of manually watching the entire video.

The analytics section can provide information about the processed content and its different components.

This allows the user to understand the video at both a high level and a detailed level.

16. EXPORT SYSTEM

The platform includes an export module.

Users can save processed information in formats such as:

PDF

Useful for reports and documentation.

JSON

Useful for structured data and future integrations.

CSV

Useful for analysis and spreadsheet-based processing.

The exported information can include:

- Transcript
- OCR results
- Insights
- Important moments
- Analysis results



17. PROJECT SERVICES

Service	Responsibility
video_service.py	Video validation and metadata
audio_service.py	Audio extraction
asr_service.py	Speech-to-text
diarization_service.py	Speaker diarization
ocr_service.py	OCR processing
video_ocr_service.py	OCR across video frames
vision_service.py	Visual analysis
analysis_service.py	Overall analysis
insights_service.py	Summary, topics and key points
timeline_service.py	Important moments and timestamps
qa_service.py	Ask Your Video
education_service.py	Educational content
export_service.py	PDF/JSON/CSV export

18. TECHNOLOGY STACK

Programming Language

Python

Used for:

- AI/ML processing
- Backend logic
- Video processing
- Data processing
- Application development
-

Frontend

Streamlit

Used to create the interactive web application.

It provides:

- Dashboard
- Upload interface
- Transcript interface



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- OCR results
- Q&A interface
- Analytics
- Export functionality

Speech AI

Faster-Whisper

Used for speech recognition and time-stamped transcription.

OCR

PaddleOCR

Used for text detection and recognition from video frames.

Computer Vision

OpenCV

Used for:

- Video reading
- Frame extraction
- Video metadata
- Image processing

Generative AI

Google Gemini

Used separately for higher-level tasks such as:

- Video Q&A
- Summarization
- Key point generation
- Education content generation
- Multimodal reasoning



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

For the AI/ML section of the project, the core ML components are Faster-Whisper, Paddle-OCR and computer vision. Gemini is the Generative AI component.

GPU

The application supports GPU acceleration using:

NVIDIA CUDA

This improves the processing speed of AI models such as Faster-Whisper.

19.SOCIAL / INDUSTRY RELEVANCE

The platform can be applied to multiple domains.

Education

Teachers and students can use the system to:

- Analyze lectures
- Generate notes
- Create quizzes
- Search lecture content
- Find important topics

Meetings

Organizations can use it to:

- Transcribe meetings
- Identify speakers
- Search conversations
- Find important discussions
- Generate summaries

Training

Training videos can be converted into searchable knowledge and learning material.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Media

Journalists and media teams can search large video archives for specific statements or topics.

Research

Researchers can process long interviews, lectures and recorded discussions.

Surveillance and Monitoring

Visual and audio analysis can help identify important events in recorded video.

INNOVATION COMPONENT

The main innovation of the project is not a single AI model.

Instead, the innovation lies in **combining multiple AI/ML capabilities into one modular video intelligence platform.**

The project uses:

- Faster-Whisper for speech
- Pyannote for speakers
- PaddleOCR for text
- Computer vision for visual information
- Modular Python services for orchestration

This provides greater control over individual components and makes the architecture customization.

The project also has a strong **education-oriented work-flow**, allowing video content to be transformed into notes, MCQs, quizzes and flashcards.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

20. LEARNING AND REFLECTION

20.1 Learning Speech Processing

One of the major technical learnings was understanding how speech recognition works in a real application.

The project involved:

- Audio extraction
- Audio preprocessing
- Speech recognition
- Timestamp generation
- GPU inference

Working with Faster-Whisper provided practical experience with speech AI.

20.2 Learning Computer Vision

Working with video frames helped in understanding that video processing is not only about playing a video.

Frames can be individually processed to extract:

- Text
- Objects
- People
- Visual information

20.3 Learning OCR

PaddleOCR provided practical experience with extracting information from images.

A major learning was that OCR accuracy depends on factors such as:

- Image quality
- Resolution
- Text size
- Orientation
- Lighting
- Frame quality



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

20.4 Learning Multimodal Processing

The biggest conceptual learning was understanding that video contains multiple information modalities.

Audio → Speech
Frames → Visual information
Frames → Text
Audio → Speakers

Combining these sources produces much richer information than analyzing only one modality.

20.5 Learning Streamlit

Developing the application using Streamlit provided practical experience with:

- Session state
- Modular UI
- File uploads
- Dynamic interfaces
- Dashboard design
- Export functionality

20.6 Learning GPU Processing

The project also provided practical experience with GPU-based inference.

The system was configured to use NVIDIA CUDA where supported.

This demonstrated the difference between CPU and GPU-based AI processing for computationally expensive models.

CONCLUSION

The **AI Video Intelligence Platform** demonstrates how Artificial Intelligence and Machine Learning can transform raw video into structured and useful knowledge. The project provides a complete end-to-end pipeline for understanding video content. The major technical achievement is the integration of multiple AI/ML components into a modular application rather than treating video as only an audio or visual medium.

“Instead of watching the entire video to find information, the AI Video Intelligence Platform allows users to let AI process, understand, search and extract knowledge from the video.”



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)